Cross-Age Face Verification from Mined Same-Person Social-Media Pairs: A Naturally-Supervised Longitudinal Signal

Anonymized for Double-Blind Review

***Abstract*—Cross-age face verification—deciding whether two photographs taken years apart depict the same person—is hard, and longitudinal training data are scarce. We ask a *data-centric* question of weakly-supervised representation learning: can the supervision for age invariance be obtained without a manually labeled longitudinal dataset? We show that multi-photo "then/now" social-media posts supply weak *same-person* longitudinal pairs with no manual identity/age labels (a large-language model parses captions and embedding clustering forms persons; we audit these components, including a small manual validation), and that fine-tuning a deliberately weak recognizer on them induces *transferable, targeted* large-gap robustness. On the external benchmark FG-NET, large-gap (≥25-year) ROC-AUC rises from 0.736 to 0.848 (+0.112 ± 0.001 over three seeds; DeLong $p < 10^{-19}$), at no measurable cost on easy benchmarks (LFW accuracy +0.004, LFW TAR@FAR=0.1% +0.002; AgeDB-30 −0.008)—a gain concentrated on the large-gap regime rather than a trade-off. We further show that the standard *random-impostor* protocol understates this task: at a 25-year gap a person is less similar to themselves than to the most similar same-age stranger (median ArcFace cosine 0.232 vs. 0.289), and a frozen state-of-the-art recognizer (AdaFace IR-101) falls from 0.961 to 0.376 [0.356, 0.394] ROC-AUC—*below chance*—once impostors are look-alikes rather than random. Our central, bounded claim is that, within this weak/medium-backbone attribution protocol, the *data source*—not the choice among the objectives we test—accounts for most of the gain: a pair-contrastive objective matches an ArcFace-margin one (0.848 vs. 0.851). The gain is driven by the same-person pairing (explicit caption age anchors add only +0.009) and is mechanistically tied to removing an age shortcut. Supporting evidence in the paper: superiority over a real re-aging model under a gap-parity control; bidirectional transfer to an independent non-VK platform (Reddit); data efficiency (≈96% of the gain at 10% of identities); an apparent-demographic audit; calibration; and a curation finding that about half of the raw "positives" were near-duplicate frames inflating internal metrics.**

***Index Terms*—Cross-age face recognition, age-invariant representation learning, weakly-supervised learning, transfer learning, dataset curation, shortcut learning, data efficiency, fairness, calibration.**

I. Introduction

CROSS-AGE face verification arises in archival retrieval, account recovery, and authorized search for missing persons. The challenge lies not in discriminating among arbitrary individuals but in distinguishing *the same person across age* from *different people of similar age and appearance*. Annotating longitudinal identity pairs (one individual across many years) is costly, and such data remain scarce, which constrains supervised approaches; recent work continues to report weak cross-dataset generalization and a shortage of high-quality longitudinal data.

Hypothesis. A multi-photo "then/now" post showing one person at different ages yields $\binom{n}{2}$ positive identity pairs without manual annotation; a caption stating ages adds a secondary age anchor. The same-person pairing is the operative signal (an ablation later isolates the age anchor at a marginal +0.009); this is a scalable, naturally-supervised signal of cross-age identity, and fine-tuning a weak recognizer on it should transfer to external benchmarks.

Contributions (bounded). (1) A weakly/naturally-supervised (no manual id/age labels) longitudinal signal mined from social posts—same-person identity pairs, with caption age anchors only a secondary cue. (2) A rigorous *attribution protocol* (one weak trainable backbone, frozen vs. fine-tuned, external evaluation) isolating the contribution of *data* from network capacity. (3) Mutually reinforcing controls (loss family, architecture, source community, training objective, real-vs-synthetic, shortcut diagnosis, and cross-platform transfer to an independent non-VK source) supporting the bounded claim that, within this protocol and across the objectives tested, the data contributes more to the large-gap gain than the choice among loss families. (4) A curation pipeline with an audit of its automatic components, and the finding that naive positive counts are inflated about 2× by near-duplicate frames. (5) Applied calibration and an apparent-demographic fairness audit with confidence intervals.

We emphasize that the novelty is data-centric rather than architectural—a new *origin of supervision* for a representation-learning system—and should be evaluated accordingly. From a learning-systems standpoint, the central question is where the cross-age learning signal originates—the *supervision source* versus network capacity and the training objective—and how the resulting transferable representation scales with the amount of naturally occurring weak supervision; *shortcut learning* and *data efficiency* are the two lenses through which we explain the effect.

II. Related Work and Positioning

Three lines of prior work are most relevant. *Discriminative age-invariant FR on labeled data:* OE-CNN (orthogonal embedding decomposition) [1], DAL (decorrelated adversarial

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Author names, affiliations, contact information, and code/data links are provided on the separate title page and are withheld here for double-blind peer review.

learning) [2], and MTLFace (multi-task recognition, age synthesis and attention decomposition) [3] achieve strong accuracy but are trained on datasets with *manual identity and age labels* (CACD [4], MORPH [5], AgeDB [6], FG-NET [7]). *Cross-age contrastive with synthetic samples:* CACon [8] adds a synthesized cross-age sample and a triplet loss; OrdCon [9] uses order-enhanced contrastive learning for generalized age features. *Synthetic aging for robustness:* systematic studies show synthetic aging helps only partially [10]. Generic self-supervised FR (SimCLR-style augmentation [11]) lacks identity anchors across time. We instead mine the structure "one post = one person at different ages" as a label-free positive-pair generator. Table I positions us against these methods.

We report on the canonical cross-age benchmarks of these methods (AgeDB-30, CALFW [12], FG-NET; CALFW was designed as a harder, age-gapped LFW [13]). The goal is *not* to surpass the absolute state of the art (we use a weak backbone for clean attribution) but to quantify the magnitude and transferability of the gain from a naturally-supervised source; Sec. V-F trains the canonical SOTA objective on our data directly.

III. Data: Naturally-Anchored Longitudinal Pairs

Source. Two public "then/now" multi-photo communities on VK (one platform, Russian-language context). Faces are detected and aligned with a 5-point ArcFace template (InsightFace RetinaFace [14], `buffalo_l`). A *usable* face passes detection-score, minimum-size, blur, and single-target checks (Sec. IX). The raw mined corpus is summarized in Table II (left).

A. Age Extraction

Captions are parsed by regular expressions (Russian patterns plus the slash format "6/18/21", one age per photo) and by a large language model (GigaChat). Across all 33,697 multi-photo captions (Table III) the LLM broadens and regularizes age extraction relative to regex—it avoids reading calendar years or gaps as ages and recovers second ages—disagreeing on 27.2% of posts, overwhelmingly LLM-favoring; on a 100-caption manual gold subset the LLM matched the human reading on 89% vs. regex 71% (supplementary material). We do not treat LLM output as ground truth: downstream, explicit age anchors contribute only marginally (Sec. V-A).

B. Persons and Leakage-Safe Split

A naive "one post = one identity" undercounts: a person recurs across posts. We cluster post-groups into persons by embedding similarity (cosine ≥ 0.85), giving 21,848 persons from 27,487 post-groups ($\approx$20% recurrence), and split *by person* (not by post). Manual inspection of high-cosine pairs showed cosine does not separate look-alikes from same-person in the 0.55–0.8 band, so the merge threshold is deliberately conservative (0.85).

C. Auditing the Automatic Supervision

The automatic supervision components are potential hidden label noise, audited three ways: LLM-vs-regex age agreement (Table III); manual inspection of high-cosine pair identity; and within-person gender consistency as an inexpensive noise detector (2,919 multi-photo groups with consistency <0.6 flag probable multi-person). A *small* manual validation (one annotator) *sanity-checks* the automatic pipeline (Table IV): it *supports*, but—with a single annotator and no inter-annotator κ—does not *certify* the supervision; a large multi-annotator audit remains beyond our resources, so residual noise is acknowledged as a limitation (Sec. VII).

D. Curation Pipeline and Its Largest Finding

Three steps: (i) *apparent gender/age metadata* (estimator) per face, aggregated per person—the corpus is 76.7% apparent-female / 23.3% apparent-male; (ii) *LLM group-integrity validation* (Table V)—person-clustering had already separated most multi-person posts, so only 421 person-groups are flagged noisy; (iii) *near-duplicate dedup* within each person (cosine $\geq$ 0.97), removing 8,190 faces ($\approx$17%).

Key finding. Because the number of positives is quadratic in the number of faces per group, removing near-duplicates reduced the positive count from 65,897 to 31,865 (-52%): **approximately half of the raw "positives" were duplicate frames** (a single shoot, burst, or repost) that had inflated the internal metrics. After curation, the frozen baseline on the internal test decreases (our.overall 0.856 $\rightarrow$ 0.836; our.25+ 0.705 $\rightarrow$ 0.640), exposing a harder and more representative benchmark. The external benchmarks are unaffected. This constitutes a transferable methodological observation for mining longitudinal pairs from social media.

Threshold robustness. The -52% reduction is not an artifact of the 0.97 cutoff. Re-running the near-duplicate dedup on the pre-curation groups at cosine thresholds 0.93–0.99 (Table VI) leaves 31.3k–32.4k positives across 0.93–0.97 (a stable $\approx$52% reduction) and degrades gracefully only at the very strict 0.99 (-41%): the near-duplicates are dominated by near-identical frames (cos $\approx$ 1), so the finding does not hinge on the exact cutoff.

E. Dataset Datasheet (Summary)

Provenance, rights and governance fields are stored per item (source, origin URL, collection date, license/consent status, allowed use, retention policy, deletion status, review status). Collection: public VK walls. Intended use: research on consent-based cross-age matching. Out of scope: mass identification, surveillance, de-anonymization. Minors: the child$\leftrightarrow$adult regime is in scope only under an explicit legal/ethical framework; per-item review status is tracked and deletion-on-request is supported. A full datasheet [15] accompanies the release.

IV. Method, Protocol and Statistics

Attribution protocol. Comparing an adapter on top of a *frozen* strong ArcFace against frozen ArcFace is invalid

TABLE I
POSITIONING VS. PRIOR WORK.

Work	Supervision	Mechanism	Our difference
OE-CNN (2018) [1]	labeled id+age	orthogonal feature decomposition	no manual id/age labels, not decomposition
DAL (2019) [2]	labeled id+age	adversarial id/age factorization	mined pairs + simpler objective
MTLFace (2021) [3]	labeled id+age	recognition + age synthesis (multi-task)	weaker pipeline; novel supervision origin
CACon (2023) [8]	semi-sup. + synthetic	contrastive on synthesized cross-age	real mined pairs > synthetic aging (Sec. V-B)
Synthetic Ageing (2024) [10]	synthetic	train on aged images	we make *real* mining the thesis
OrdCon (2025) [9]	age-ordered contrastive	order-enhanced features	mine natural pairs, no explicit age order

TABLE II
DATASET SCALE (RAW MINED) AND AFTER CURATION (SEC. III-D).

Stage	Posts	Photos	Usable faces	Persons (groups)	Positive pairs
Raw mined (2 communities)	44,609	84,147	49,606	21,848	65,897
After curation	—	—	**40,586**	**21,427**	**31,865**

TABLE III
AGE-EXTRACTION AUDIT: LLM VS. REGEX (33,697 CAPTIONED MULTI-PHOTO POSTS).

Category	n	Share
Agree (same ages or both empty)	24,531	72.8%
LLM filled (regex missed)	3,656	10.8%
Age mismatch	4,658	13.8%
LLM missed (regex found)	852	2.5%

TABLE IV
SMALL MANUAL VALIDATION (ONE ANNOTATOR; A SANITY CHECK, *not* A CERTIFIED AUDIT—NO INTER-ANNOTATOR κ). SAMPLES ARE DRAWN FROM THE CURATED CORPUS AND WERE NOT USED TO TUNE THRESHOLDS OR FILTERING RULES.

Component	n	Sampling	Human-checked result
Age extraction	100	strat. random	LLM 89% vs. regex 71% exact
Group integrity	100	random+risk	single/multi acc. 0.96
Positive pairs	200	random	same-person prec. 0.96
Cluster merges	100	near-thresh.	merge prec. 0.95
Near-dup removals	100	random	duplicate prec. 0.93

TABLE V
LLM GROUP-INTEGRITY CLASSIFICATION (33,697 POSTS).

Category	n	Share
Single (one person across ages)	31,275	92.8%
Multi-person (different people)	2,034	6.0%
Unknown	336	1.0%
Meme / collage	52	0.2%

(the adapter merely deforms near-perfect embeddings). Instead we use *one weak trainable backbone* (FaceNet InceptionResnetV1 [16], CASIA-WebFace) and compare (a) frozen → benchmark metric vs. (b) soft fine-tuning of the head on our pairs → benchmark metric. Training uses a margin contrastive loss on aligned crops. *Evaluation is on external benchmarks*

TABLE VI
DEDUP-THRESHOLD SENSITIVITY: NEAR-DUPLICATE REMOVAL ALONE, ON THE PRE-CURATION GROUPS (THE SEPARATE GROUP-INTEGRITY PRUNE REMOVES A FURTHER ≈500 POSITIVES TO REACH THE HEADLINE 31,865). STABLE ACROSS 0.93–0.97.

Dedup cos	Near-dup faces	Positives	Reduction
0.93	8,653	31,304	−52.5%
0.95	8,564	31,611	−52.0%
0.97 (used)	8,304	32,360	−50.9%
0.99	6,274	38,565	−41.5%

(LFW, AgeDB-30, CALFW, FG-NET) for a clean attribution of transfer; the internal test (`our.*`) is a secondary in-domain diagnostic. Backbone strength is varied (Sec. V-C) to map where the data helps. The design trades absolute performance for *causal attribution*.

Backbones. FaceNet (weak); ArcFace iResNet-50 [17] (CASIA-FaceV5; weak, different architecture); AdaFace IR-50/IR-101 [18] (WebFace; strong; a clean depth control); ArcFace iResNet-100 (strong).

Benchmarks and metrics. LFW (10-fold accuracy), AgeDB-30 and CALFW (aligned pairs; ROC-AUC and 10-fold accuracy), FG-NET (ROC-AUC overall and on the large-gap ≥25-year subset). We deliberately do not rest the headline on FG-NET alone (small, old, partial detector coverage): AgeDB-30 and CALFW show that the large-gap gain comes without a catastrophic collapse on standard cross-age benchmarks (AgeDB-30 decreases slightly, CALFW is nearly unchanged), confirming that the effect is concentrated on the FG-NET large-gap subset rather than uniform across benchmarks. We distinguish *threshold-free* metrics (ROC-AUC) from *threshold-based* ones (accuracy, TAR@FAR). LFW is evaluated on *aligned* crops: the pair images are re-detected with RetinaFace and warped by the same 5-point `norm_crop` used for our own crops (0.36% detector misses). This matters: the commonly used `sklearn` loader returns unaligned images cropped to 125×94, on which margin-trained recognizers collapse (frozen

ArcFace r100 scores 0.83 there while scoring 0.98 on the harder AgeDB-30). Absolute LFW values under our alignment sit slightly below published ones and should not be compared to leaderboards; frozen and fine-tuned models are measured identically, so the reported differences are valid.

Statistics. Our *primary endpoint* is the FG-NET large-gap ($\geq$25-year) ROC-AUC, with the null hypothesis that fine-tuning on the mined pairs does not improve it over the frozen baseline (gain ≤ 0); AgeDB-30, CALFW and the internal cross-age test are secondary endpoints, and easy LFW accuracy is monitored as a forgetting control. The headline gain is mean $\pm$ std over three seeds (seed variance captures training stochasticity, not full evaluation uncertainty). The fairness audit uses a paired percentile bootstrap (1000 resamples, resampling pairs and recomputing both models on the same resample) for the *gain*, accounting for frozen/tuned correlation. Table VIII reports bootstrap CIs, EER and TAR@FAR for the key benchmark comparisons (the internal test additionally uses an identity-level, person-resampling bootstrap), and a Holm correction is applied across the stratified fairness tests.

Curated dataset. All headline experiments are on the curated data of Sec. III-D: 21,427 person groups, 40,586 faces, 31,865 positives; split train 22,179 / val 4,579 / test 5,107 with balanced negatives per split.

V. Results

A. Main Effect and Supervision-Source Ablation

The weak FaceNet, softly fine-tuned on our pairs, raises FG-NET large-gap ROC-AUC **0.736** $\rightarrow$ **0.848** ($+0.112 \pm 0.001$), while easy LFW does not degrade at all ($+0.004$) and AgeDB-30/CALFW move negligibly (Table VII). An ablation isolates the source: same-post pairs use no manual labels—neither identity nor age—and deliver almost all of the gain; adding explicit age anchors contributes only $+$**0.009** on the 25+ bucket (pre-clean ablation). The method is therefore *weakly supervised by naturally occurring same-post co-occurrence* (not by manual identity/age labels), with age supervision contributing only marginally.

The gain is statistically robust (std ≤ 0.003) and survives curation; the internal cross-age gain even *grows* on the harder curated test (our.25+ $+0.195$) because curation lowers the frozen baseline to 0.640 while +pairs recovers to 0.835.

Bootstrap confidence intervals (seed 42) confirm the primary endpoint: FG-NET large-gap rises from 0.736 [0.70, 0.78] to 0.848 [0.82, 0.88] with *non-overlapping* 95% CIs; the equal-error rate **nearly halves** (0.335→0.219) and TAR@FAR=1% **nearly doubles** (0.21→0.39). On the internal 25+ test the gain holds under the more conservative *identity-level* bootstrap (resampling persons, not pairs): +pairs 0.838 [0.80, 0.88] vs. frozen 0.640 [0.59, 0.69], an interval wider than the pair-level one as expected when pairs share identities. Operating points and intervals for all benchmarks are in Table VIII. Because bootstrap-CI overlap ignores the samples the two models share, we confirm the primary endpoint with DeLong's test for two *correlated* ROC curves [19]: on FG-NET large-gap $z = 9.3$, $p = 1.4 \times 10^{-20}$ (ΔAUC 95% CI $[+0.088, +0.135]$), and on the internal 25+ test $p = 1.7 \times 10^{-34}$; a 10^4-fold paired permutation test agrees ($p < 10^{-4}$). The same test confirms the gain is *targeted*: AgeDB-30 shows a small but significant decrease (-0.008, $p = 4 \times 10^{-6}$) and CALFW is statistically unchanged ($p = 0.95$).

B. Real Pairs Outperform Synthetic Aging

Replacing real positives with synthetically aged versions—a domain proxy and a real re-aging network (FRAN [20])—underperforms real pairs on the curated data, and at FRAN's default *wide-gap* setting even *degrades* below frozen: FG-NET large-gap 0.727 (**below frozen 0.736**) and our.25+ 0.549 (below frozen 0.640), whereas real pairs yield $+0.112$ / $+0.198$ (Table IX). Identity-preserving aging reproduces texture artifacts rather than genuine longitudinal variation (pose, camera, era). A native-resolution control rules out a crop-resolution artifact: re-cropping a train subset at 512 px directly from the raw images (vs. our stored 112-px crops, which FRAN upscales internally) and re-aging **leaves the result unchanged**—FG-NET large-gap 0.771 vs. 0.778 and our.25+ 0.671 vs. 0.670 on this 499-face subset—so synthetic aging fails to match real pairs regardless of source resolution. A gap-parity control sharpens this: FRAN ages to a wide 33–70-year gap whereas real pairs have a median gap of 9 years, and matching the synthetic gap to the real distribution lifts FRAN to **0.759** (our.25+ 0.622)—a *modest* positive just above frozen, in line with small synthetic-aging gains reported elsewhere [10]—yet real pairs (0.848) still lead by $+0.089$. The sub-frozen figure was thus partly a gap-mismatch artifact: real mined pairs beat even a *gap-matched* re-aging model.

C. Headroom Curve and the Strong-Backbone Regime

The gain reproduces on FaceNet, ArcFace iResNet, and AdaFace IR-50/IR-101 (three loss families, two architectures) and *monotonically decreases with frozen strength* (Table XI, Fig. 1); a clean depth control (AdaFace IR-50 $\rightarrow$ IR-101) confirms it. Strong saturated backbones do not improve and forget slightly under fine-tuning; a gentler learning rate (1e-5) approximately halves the forgetting but does not convert it into a gain. The contribution is therefore specific to weak and medium backbones with cross-age headroom—a limitation we state explicitly (Sec. VII).

The random-impostor protocol understates the task. Our test negatives are random cross-identity pairs, which modern backbones separate trivially: frozen AdaFace IR-101 reaches `our.25+` $= 0.968$. That validates by measurement, rather than merely asserts, the attribution argument of Sec. IV: under this protocol strong embeddings really are near-perfect, so an adapter on top of them would have nothing to attribute. The reason is visible in the raw similarities. Median ArcFace cosine is 0.456 for positives below 10 years, 0.307 at 10–25 years and 0.232 at 25+, against 0.012 for random negatives—but **0.289** for the most similar same-gender, same-apparent-age impostor. *At a 25-year gap a person is less similar to themselves than to the most similar same-age stranger.* Re-mining the negatives accordingly (same apparent gender, $|\Delta\text{age}| \leq 5$ years, ranked by an independent miner outside the evaluated set) collapses every modern recognizer we test (Table X): frozen IR-101 falls

TABLE VII
MAIN RESULT: WEAK FACENET FROZEN VS. +PAIRS, THREE SEEDS, CURATED DATA. PRE-CLEAN GAIN SHOWN FOR REFERENCE.

Metric	Frozen	+pairs (mean ± std)	Gain	(Pre-clean gain)
FG-NET large-gap	0.7361	0.8484 ± 0.0007	+**0.1124**	+0.1215
FG-NET ROC	0.8955	0.9230 ± 0.0006	+0.0275	+0.0277
our.25+ (internal)	0.6403	0.8348 ± 0.0027	+**0.1946**	+0.1677
our.overall (internal)	0.8363	0.9163 ± 0.0009	+0.0801	+0.0764
LFW accuracy	0.9640	0.9679 ± 0.0002	+0.0039	n/a
AgeDB-30 ROC	0.9530	0.9462 ± 0.0013	−0.0068	−0.0155
CALFW ROC	0.9482	0.9489 ± 0.0014	+0.0007	−0.0030

TABLE VIII
OPERATING POINTS AND 95% BOOTSTRAP CIS (FACENET FROZEN VS. +PAIRS, SEED 42). ALL VALUES ARE ROC-AUC UNLESS NOTED; LFW IS SHOWN AS ROC-AUC FOR CONSISTENCY (ITS 10-FOLD ACCURACY IS IN TABLE VII). EXTERNAL CIS ARE PAIR-LEVEL; THE INTERNAL TEST ADDITIONALLY USES AN IDENTITY-LEVEL BOOTSTRAP (TEXT).

Benchmark	Frozen AUC [95% CI]	+pairs AUC [95% CI]	EER (fr.→+p.)	TAR@1%	TAR@0.1%
FG-NET large-gap	0.736 [0.70, 0.78]	**0.848 [0.82, 0.88]**	0.335→0.219	0.209→0.386	0.074→0.154
FG-NET overall	0.896 [0.89, 0.90]	0.922 [0.92, 0.93]	0.186→0.151	0.502→0.562	0.315→0.308
AgeDB-30	0.953 [0.95, 0.96]	0.945 [0.94, 0.95]	0.115→0.126	0.525→0.510	0.285→0.309
CALFW	0.948 [0.94, 0.95]	0.948 [0.94, 0.95]	0.116→0.117	0.580→0.624	0.213→0.320
LFW (AUC)	0.983 [0.98, 0.99]	0.986 [0.98, 0.99]	0.038→0.037	0.939→0.940	0.814→0.816

TABLE IX
REAL VS. SYNTHETIC POSITIVES (FACENET, CURATED DATA). FRAN-GM MATCHES THE SYNTHETIC AGE-GAP TO THE REAL-PAIR DISTRIBUTION (MEDIAN 9 Y); THE CRUDER DOMAIN PROXY REACHES 0.761 / 0.594.

Metric	Frozen	+real	syn-FRAN	FRAN-gm
FG-NET large-gap	0.736	**0.848**	0.727	0.759
our.25+	0.640	**0.838**	0.549	0.622

from 0.961 to **0.376** [0.356, 0.394] at rank-1, *significantly below chance*. This is not a labeling artifact: the mined impostors are 0.00% same-person under the identity clustering, and the benchmark carries zero identity leakage across splits, zero same-person negatives (0/31 865), 99.1% LLM-verified single-person groups among those surviving into positives (92.8% before curation removes multi-person groups) and 0.00% near-duplicates (Sec. III-D). A second miner from a different loss family reproduces the curve to within 0.03 and preserves the model ordering at every level, so the difficulty axis belongs to the task, not to the miner. Our mined pairs still help a weak backbone across the whole curve (+0.20 to +0.28 on the 25+ slice) and the distance to frozen SOTA narrows at the hard end (−0.145 at random vs. −0.070 at rank-1, paired bootstrap), but it does not close. The correct reading of Table XI is therefore that strong backbones are not improved by *our* supervision—not that they lack cross-age headroom, and not that the task is solved.

D. Data-Scaling Law

Varying the fraction of training identities (val/test fixed), the gain saturates early: **≈96% of the full FG-NET large-gap gain is reached with only 10% of identities** (≈1,500), reaching a plateau from 25% (Table XII, Fig. 2; each point is mean ± std over three seeds); the slight dip at full data (0.50→1.00) lies within seed variance—the curve is flat, not monotonic, beyond ≈10%. The signal is thus data-efficient; further gains require *harder* data (larger gaps, hard positives) rather than merely a greater volume.

TABLE X
LOOK-ALIKE IMPOSTORS BREAK MODERN FACE RECOGNITION (25+ ROC-AUC; RANK = IMPOSTOR RANK UNDER AN INDEPENDENT MINER). *Top:* FROZEN RECOGNIZERS ON THE *full* 25+ SET ($n_{\text{POS}} = 1208$, $n_{\text{NEG}} = 31\,590$), VALID BECAUSE NO FROZEN MODEL HAS SEEN ANY OF OUR DATA. *Bottom:* THE EFFECT OF OUR SUPERVISION, NECESSARILY TEST-ONLY ($n_{\text{POS}} = 141$, $n_{\text{NEG}} = 4762$). THE BLOCKS USE DIFFERENT IMPOSTOR POOLS AND ARE *not* COMPARABLE ACROSS BLOCKS: RANK-k DIFFICULTY GROWS WITH POOL SIZE.

Frozen recognizers, full 25+ set			
Negatives	AdaFace IR-101	ArcFace r100	AdaFace IR-50
random	0.961	0.941	0.908
rank-50	0.727	0.677	0.560
rank-10	0.603	0.565	0.437
rank-1	**0.376**	**0.362**	**0.258**
Our supervision, test split only			
Negatives	FaceNet frozen	FaceNet +pairs	Δ
random	0.568	0.811	+0.243
rank-50	0.381	0.657	+0.276
rank-10	0.305	0.562	+0.257
rank-1	0.203	0.404	+0.201

E. Mechanism — the Age Shortcut

Under age-matched negatives (same age bucket, different people), frozen accuracy on the internal 25+ test falls from 0.640 to 0.517 (**near the chance level of 0.5**), and real pairs restore it to **0.838** (Fig. 3)—direct evidence that frozen

TABLE XI
HEADROOM: FG-NET LARGE-GAP BY BACKBONE STRENGTH (CURATED DATA, LR 3E-5).

Backbone (strength)	Frozen	+pairs	Δ large-gap	Δ our.25+
FaceNet (weak)	0.736	0.848	+**0.112**	+0.198
ArcFace r50 (weak, diff. arch.)	0.764	0.805	+0.041	+0.133
AdaFace IR-50 (strong)	0.917	0.924	+0.007	+0.004
ArcFace r100 (strong)	0.954	0.915	−0.039	−0.002
AdaFace IR-101 (strong)	0.957	0.931	−0.026	−0.006

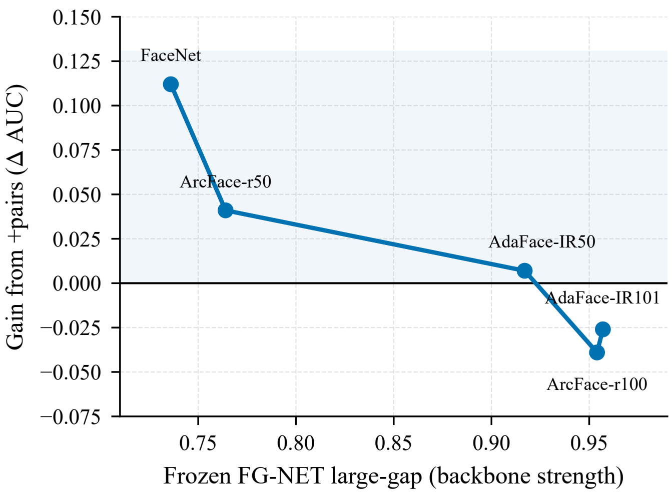

Fig. 1. Headroom curve: the +pairs gain on FG-NET large-gap shrinks monotonically as the frozen backbone strengthens, crossing zero for strong saturated models.

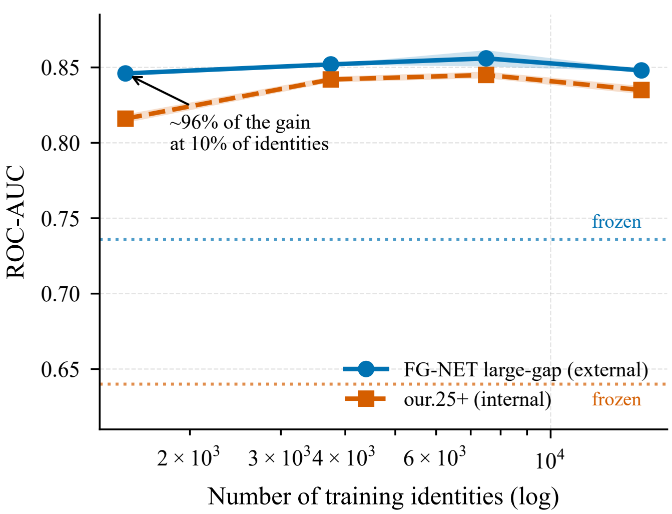

Fig. 2. Data-scaling law: external (FG-NET large-gap) and internal (our.25+) ROC-AUC vs. the number of training identities (log scale). Roughly 96% of the gain is reached at 10% of identities; shaded bands are ±1 std over three seeds and dashed lines are the frozen baselines.

TABLE XII
DATA-SCALING LAW (FACENET, CURATED; VAL/TEST FIXED; MEAN ± STD OVER THREE SEEDS).

Train fraction	# identities	FG-NET large-gap	our.25+
frozen	0	0.736	0.640
0.10	≈1,500	0.846 ± 0.002	0.816 ± 0.003
0.25	≈3,750	0.852 ± 0.000	0.842 ± 0.002
0.50	≈7,499	0.856 ± 0.005	0.845 ± 0.002
1.00	≈14,998	0.848 ± 0.001	0.835 ± 0.003

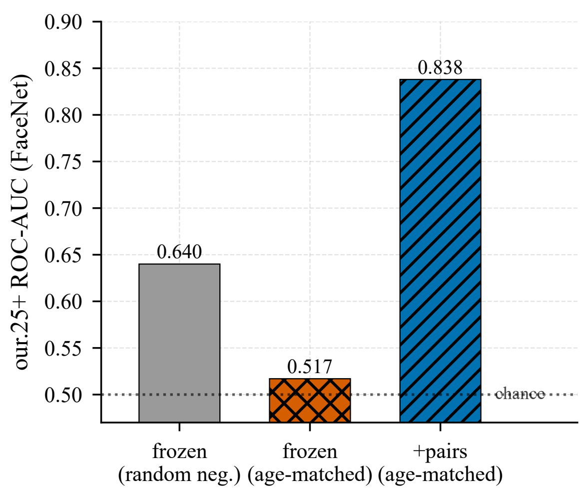

Fig. 3. The age shortcut: on an identity-only test (age-matched negatives), the frozen model falls to near-chance; fine-tuning on our pairs restores discrimination by identity.

TABLE XIII
AGE-LEAKAGE LINEAR PROBE (CURATED DATA; CHANCE = 0.25).

Model	Age-probe bal. acc. ↓	Identity ROC-AUC ↑
frozen	0.651 ± 0.019	0.836
+pairs	0.629 ± 0.012	0.916
+pairs+disentangle	0.632 ± 0.017	0.918

models discriminate cross-age pairs largely by *age*, whereas our pairs induce discrimination by *identity*. A linear probe refines this *representationally* (Table XIII): apparent age stays well decodable from the identity embedding for *all* models (≈0.63–0.65 balanced accuracy ≫ 0.25 chance) and changes only slightly, while identity AUC rises sharply—so the method learns an invariant *metric* (the cosine stops relying on the age axis), not an age-free representation; explicit adversarial disentanglement reduces leakage no further.

F. Objective Comparison: Training the SOTA Objective on Our Data

We train the canonical modern-FR margin objectives—ArcFace-, CosFace- and SphereFace-margin identity classifica-

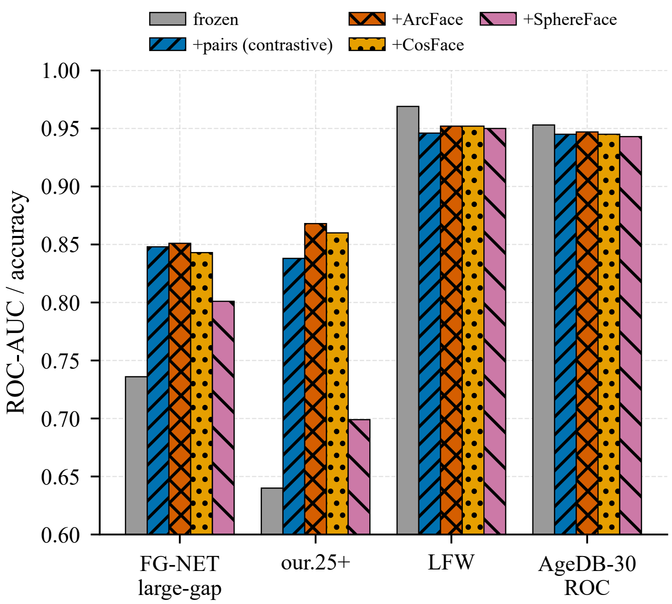

Fig. 4. Objective comparison: contrastive, ArcFace and CosFace coincide on cross-age transfer; SphereFace (hardest to optimize, no λ-annealing) trails. ArcFace/CosFace forget slightly less on easy benchmarks.

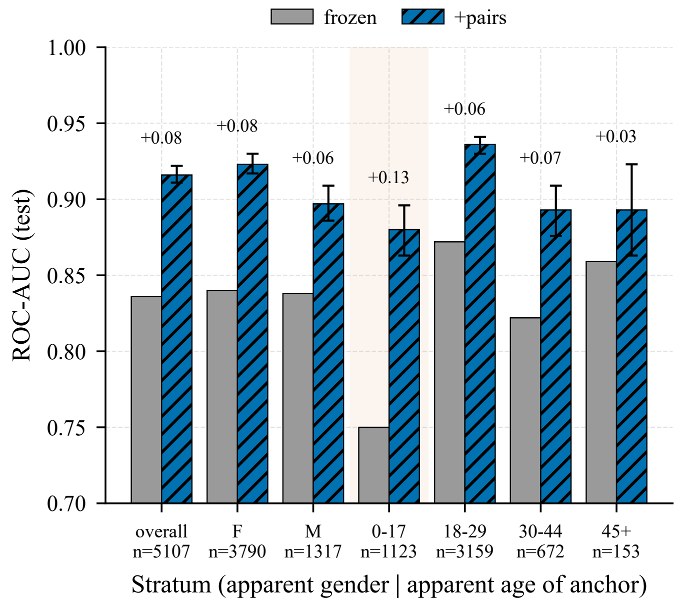

Fig. 5. Apparent-demographic audit: frozen vs. +pairs ROC-AUC across apparent gender/age strata. Every stratum improves; the largest gain is for the youngest (0–17) group, where the frozen baseline is weakest. Error bars are the 95% paired-bootstrap CI of the gain and n is positive pairs per stratum (the sparse 45+ stratum, n=153, has a correspondingly wide interval).

TABLE XIV
OBJECTIVE COMPARISON (FACENET, CURATED DATA): CONTRASTIVE, ARCFACE AND COSFACE COINCIDE ON THE FG-NET LARGE-GAP PRIMARY ENDPOINT; SPHEREFACE (NO λ-ANNEALING) TRAILS ON OUR SPARSE FEW-SHOT IDENTITIES.

Objective	FG-NET l.g.	our.25+	LFW	AgeDB-30
Frozen	0.736	0.640	0.969	0.953
+pairs (contrastive)	0.848	0.838	0.946	0.945
+ArcFace	**0.851**	0.868	0.952	0.947
+ArcFace, sub-center	0.851	0.869	0.951	0.947
+CosFace	0.843	0.860	0.952	0.945
+SphereFace	0.801	0.699	0.950	0.943

tion [17] (the core of OE-CNN/MTLFace)—on our naturally-supervised identities (9,204 classes), same weak backbone, scope and protocol. On the FG-NET large-gap primary endpoint the contrastive, ArcFace and CosFace objectives **coincide** (0.848 / 0.851 / 0.843, within $\pm$0.006; Table XIV, Fig. 4); only SphereFace—the hardest margin to optimize, here without λ-annealing on 2–3-shot identities—trails (0.801). We draw the bounded conclusion that, within this protocol and among the objectives tested, the data source accounts for more of the gain than the objective choice—not that the objective is irrelevant in general. ArcFace/CosFace additionally *forget less* on easy benchmarks—a practical recommendation. A *noise-robust* comparator—sub-center ArcFace [21], which keeps K=3 centroids per identity so that label-noisy or low-quality crops route to off-centers—lands at the *same* large-gap 0.851 (our.25+ 0.869); explicit label-noise handling thus adds nothing over a plain margin on our *curated* mined pairs, consistent with deduplication having already removed the dominant noise source. The supervision source, not noise-robust loss machinery, drives the gain.

G. Apparent-Demographic Audit

Stratifying by *apparent* gender/age of the pair anchor (estimated, *not* self-identified), no stratified group degrades and every gain is significant (95% paired-bootstrap CI $>$ 0; Table XV, Fig. 5). The largest gain is for the youngest group 0–17 (frozen 0.750 $\rightarrow$ +pairs 0.880, +0.130, CI not overlapping adult bands)—the method strengthens where the base model is weakest (the child$\leftrightarrow$adult regime), narrowing the apparent-demographic gap. This is corroborated on *independent, non-VK* data: on a child$\leftrightarrow$adult subset of FG-NET (one photo under age 13, the other over 25; 195 positives, 195 negatives), +pairs raises ROC-AUC from 0.684 [0.63, 0.74] to 0.813 [0.77, 0.86] (+**0.129, non-overlapping 95% CIs**), closely matching the internal +0.130. Beyond AUC, an error decomposition at a fixed operating point (per-model global threshold at 1% FMR) shows a genuine reduction in matching errors where it matters: +pairs lowers the false-non-match rate in *every* apparent stratum at matched $\approx$1% FMR—0–17 **0.852$\rightarrow$0.656**, apparent-female 0.674$\rightarrow$0.509, apparent-male 0.743$\rightarrow$0.534—with no stratum regressing, and per-stratum score calibration (logistic slope 9.0–11.7) stays uniformly positive and stable; the only exception is the small 45+ stratum ($n\approx$150), whose FMR scatters to 2.1% at the global threshold (sampling noise). We deliberately avoid the word "fair" as a solved property: the source is apparent-female-skewed (76.7%), attributes are apparent (not ethnicity or legal categories), and 45+ is sparse.

H. Cross-Source Transfer

Training on one community and testing on the held-out other transfers the gain: external FG-NET large-gap +0.113 (vs. +0.112 within-source—essentially identical) and internal +0.092 on the held-out community. We frame this as transfer

TABLE XV
APPARENT GENDER/AGE STRATA (FACENET, CURATED; 95% PAIRED-BOOTSTRAP CI OF THE GAIN).

Stratum	n_{pos}	Frozen	+pairs	Gain	95% CI
overall	5,107	0.836	0.916	+0.080	[+0.075, +0.086]
apparent F	3,790	0.840	0.923	+0.084	[+0.077, +0.090]
apparent M	1,317	0.838	0.897	+0.059	[+0.048, +0.071]
age 0–17	1,123	0.750	0.880	**+0.130**	[+0.113, +0.146]
age 18–29	3,159	0.872	0.936	+0.064	[+0.058, +0.069]
age 30–44	672	0.822	0.893	+0.070	[+0.054, +0.087]
age 45+	153	0.859	0.893	+0.034	[+0.004, +0.064]

TABLE XVI
CROSS-PLATFORM TRANSFER: A VK-TRAINED MODEL ON AN INDEPENDENT REDDIT (NON-VK) THEN/NOW TEST SET (FROZEN VS. +PAIRS, TRAINED ON VK ONLY; 396 POSTS → 1,575 POSITIVE PAIRS AFTER THE MULTI-PERSON-COLLAGE FILTER).

Reddit (non-VK) test metric	Frozen	+pairs (VK-trained)
Overall ROC-AUC [95% CI]	0.718 [0.70, 0.74]	**0.749 [0.73, 0.77]**
EER	0.345	**0.328**
TAR@FAR=1%	0.271	**0.279**
Test pairs (pos / neg)	1,575 / 1,575	

between two independent sources within one platform domain; Sec. V-I extends the test to an independent non-VK platform, and we do not claim web-scale generalization.

I. Cross-Platform Transfer (Independent, Non-VK Source)

A key external check of a naturally-supervised *source* is whether it generalizes beyond the originating platform. We apply the *identical* mining protocol to an independent, non-VK community—the Reddit board r/PastAndPresentPics (same “then/now” format, same LLM parser)—and evaluate the VK-trained +pairs model (*no* Reddit data in training) on 1,575 Reddit pairs (Table XVI): overall ROC-AUC rises **0.718** → **0.749** (+0.031, near-disjoint 95% CIs), EER 0.345 → 0.328. The transfer is *bidirectional* (Table XVII): a weak backbone trained on *only* the small Reddit source—a 220-person *pilot*, not a headline claim—also helps on VK. Reddit then/now pairs are inherently cross-age, so the frozen 0.718 sits far below its easy-benchmark ceiling, mirroring the within-VK large-gap regime; the gain is smaller than within-VK (a noisier, partly collage-derived set), so we read it as confirming the *source* generalizes beyond its platform, not as a SOTA claim. The explicit ≥25-year subset is not reported separately (Reddit titles seldom state exact ages). Figure 6 places this alongside our other external evaluations: the +pairs gain stays *positive across all three*, shrinking on harder/noisier sets.

J. Calibration

Raw cosine (Platt) is mis-calibrated across gap bins—over-confident at 15–25 years (ECE 0.052); a $P(\text{same} \mid \text{cosine}, \text{age-gap})$ calibrator fixes every bin (15–25 → 0.017) and improves the overall Brier score (0.035 → 0.029)—a usable, age-aware same-identity probability.

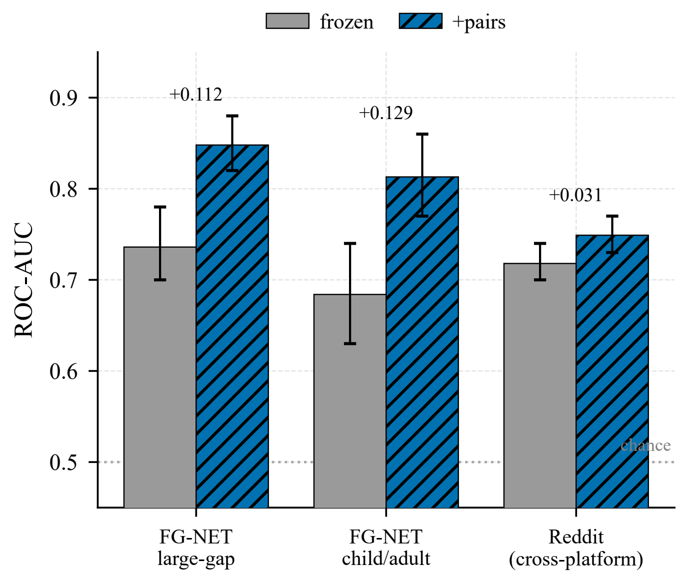

Fig. 6. External generalization. The +pairs gain over the frozen baseline (with 95% CIs) holds across three independent evaluations—FG-NET large-gap, a child↔adult FG-NET subset, and the cross-platform Reddit set—each well above chance. The gain is largest on the in-protocol FG-NET sets and smaller (but positive) on the noisier cross-platform Reddit set.

TABLE XVII
CROSS-SOURCE TRANSFER, BOTH DIRECTIONS. VK→REDDIT IS THE MAIN CROSS-PLATFORM TEST; REDDIT→VK IS A SMALL 220-PERSON *pilot* SUPPORTING SOURCE NON-SPECIFICITY, NOT A HEADLINE CLAIM (SINGLE RUN).

Train	Test	Frozen	Tuned	Δ
VK	Reddit (overall)	0.718	0.749	+0.031
Reddit	FG-NET large-gap	0.736	0.783	+0.047
Reddit	VK internal 25+	0.640	0.659	+0.019

K. Disentanglement

Explicit age removal (gradient reversal + age head, MTLFace-style) adds only a small, stable cross-age gain over +pairs (+0.006 FG-NET large-gap, +0.015 our.25+) without harming easy benchmarks; its absolute large-gap (≈0.853) matches the ArcFace objective (0.851), reinforcing that the data—not the objective or the disentangling add-on—sets the cross-age ceiling.

TABLE XVIII
HARD-NEGATIVE MINING (FACENET, CURATED DATA; EXTERNAL BENCHMARKS). +HARD-NEG IS MEAN ± STD OVER THREE SEEDS; FROZEN AND +PAIRS AS IN TABLE VII.

Metric	Frozen	+pairs	+pairs+hard-neg
FG-NET large-gap	0.736	0.848	**0.868 ± 0.004**
FG-NET ROC	0.896	0.923	0.912 ± 0.003
LFW accuracy	0.969	0.950	0.925 ± 0.005
AgeDB-30 ROC	0.953	0.946	0.911 ± 0.013
CALFW ROC	0.948	0.949	0.898 ± 0.015

L. Hard-Negative Mining

Adding the hardest cross-person negatives to fine-tuning—the top-5 most-similar other-person faces per anchor (cosine in a hard band, excluding the near-duplicate/uncertain range)—further sharpens large-gap discrimination: FG-NET large-gap rises to **0.868 ± 0.004** over three seeds (from 0.848 with random negatives), but easy benchmarks forget *more* and FG-NET overall ROC actually drops (LFW 0.969→0.925, AgeDB-30 0.953→0.911, CALFW 0.948→0.898, FG-NET ROC 0.923→0.912; Table XVIII). Hard negatives thus trade general-benchmark accuracy for large-gap separation—useful for a cross-age retrieval setting, not for a general verifier. (A secondary result, not part of the headline; the internal hard-negative test is near-chance for the frozen model by construction and is not comparable to the standard internal test.)

VI. DISCUSSION

The contribution is a scalable, naturally-supervised source of longitudinal supervision that teaches transferable cross-age robustness which synthetic aging cannot replace and explicit age supervision does not explain. The reinforcing controls—loss family, architecture, source community, objective, and a noise-robust margin—converge on a bounded conclusion: within this protocol, the *data source* accounts for more of the large-gap gain than the choice of objective. The improvement is *concentrated* on the large-gap regime rather than uniform, but it is not a trade-off: on correctly aligned LFW every operating point is flat or better (TAR@FAR=0.1% 0.814 → 0.816, EER 0.038 → 0.037; Table VIII), and CALFW improves at low FAR (+0.106 at TAR@FAR=0.1%). Only AgeDB-30 shows a small significant decrease (−0.008). An earlier version of this analysis reported substantial low-FAR degradation; that was an artifact of an unaligned LFW evaluation path (Sec. IV) and does not survive correction.

A. Scope of the Claims

To forestall over-reading, we state explicitly what the evidence does and does not establish.

Established (within this attribution protocol). (i) the mined pairs improve large-gap (≥25-year) cross-age verification on FG-NET (the primary endpoint, with non-overlapping 95% CIs) and on the internal 25+ test; (ii) the gain transfers across the two source communities and to an independent child↔adult FG-NET subset; (iii) real pairs **outperform synthetic aging**, including under native-resolution and gap-parity controls; (iv) the gain is invariant across the contrastive, ArcFace and CosFace objectives; (v) it is mechanistically tied to **removing an age shortcut**; (vi) roughly half of the raw positive pairs were near-duplicate frames.

Not established / out of scope. (a) the effect is *not* a uniform verification gain—it is concentrated on the large-gap subset, with easy benchmarks flat (Table VIII); (b) strong saturated backbones do not improve; (c) cross-platform transfer is shown on an independent non-VK platform (Reddit then/now, Sec. V-I, +0.031 overall AUC) and on a child↔adult FG-NET subset; a recognizer trained on *only* a non-VK source already transfers (Sec. V-I); a larger non-VK *training* source with multi-seed CIs remains future work; (d) we do not reproduce a full state-of-the-art pipeline, only its shared margin objective; (e) the fairness audit uses apparent attributes only and lacks a human-audited inter-annotator κ (full manual annotation is labor-intensive and beyond our resources; we report only automatic cross-checks) and a full per-cell multiple-comparison correction (release supplement).

VII. LIMITATIONS

- **Backbone dependence:** the gain concentrates in weak/medium backbones; strong models do not improve and slightly forget (gentle lr mitigates but does not reverse this). They are saturated only against random impostors—with look-alike negatives they are far from ceiling (Table X), so this bounds *our* supervision, not the task.
- **External validity:** two VK communities, one platform, one language/cultural context; apparent-female-skewed (76.7%); sparse at apparent age 45+. Transfer is shown between two VK sources, to standard external benchmarks, and to a small independent Reddit then/now test; this does not establish platform-agnostic generalization.
- **Apparent attributes only** in the fairness audit (estimated, not self-identified gender/ethnicity).
- **Naturally-supervised, not strictly label-free:** an LLM parses ages and validates group integrity; clustering forms persons—audited (Sec. III-C) but a residual hidden-noise source.
- **Statistics:** seed variance, paired bootstrap (fairness), and benchmark CIs / EER / TAR@FAR (Table VIII, including an identity-level bootstrap) are reported in-text; DeLong tests for correlated ROC curves, an FMR/FNMR error decomposition and per-stratum calibration are now reported (Sec. V-A, Sec. V-G); a full per-cell multiple-comparison correction across every benchmark×operating-point remains for the release supplement.
- **Benchmark scale and age:** we evaluate on the field-standard cross-age suite—AgeDB-30 and CALFW (6,000 pairs each) and FG-NET for the explicit ≥25-year subset—the same benchmarks used by OE-CNN/DAL/MTLFace, which fixes comparability but inherits their limited scale and age (FG-NET in particular is small with partial detector coverage, so we never rest the headline on it alone). Our curated internal test (5,107 cross-age positives with controlled negatives) is a sizeable modern complement; we additionally report a child↔adult subset of FG-NET (Sec. V-G) and

a cross-platform transfer test on an independent non-VK source (Reddit, Sec. V-I); a larger non-VK *longitudinal* set remains useful future work.

- **SOTA pipelines not reproduced end-to-end:** by design we train the *shared core* of modern AIFR—the ArcFace angular-margin objective—under our weak-backbone protocol for clean attribution, not a leaderboard comparison. Sec. V-F shows this objective matches our contrastive one on the key external metric, so the objective is not what drives the gain; the omitted method-specific add-ons (MTLFace's generative age-synthesis branch, OE-CNN's orthogonal-subspace decomposition) refine that shared objective and are orthogonal to our claim about the supervision source. Whether a full SOTA pipeline trained on naturally-supervised data closes the remaining absolute gap is a well-scoped follow-up.

VIII. Ethics, Legal Basis and Data Governance

This is a sensitive biometric study; we treat governance as a first-class component rather than as a disclaimer. Permitted use is restricted to controlled / consent-based / human-reviewed applications (rights-cleared archives and family photos, authorized humanitarian search, account recovery); the system outputs candidates for human review, never an automatic decision. This is a *research-only* study—nothing is deployed, no automated decision is made, and *no public biometric database is released* (Sec. VIII-B). Forbidden: mass identification, de-anonymization, surveillance, and any processing of minors' data without a legal basis.

A. Ethical Approval and Legal Basis

Ethical approval. The study was not placed under institutional review board oversight; IEEE policy permits an explanation in place of an approval reference, and this is ours. There is no interaction or intervention with any person: we retrospectively analyse images the subjects had themselves published on public community walls, retrieved via the platform's official API. Nobody is identified, contacted, profiled or ranked; nothing is deployed; minors are excluded as subjects; and no released artifact contains raw faces, raw posts or recoverable identifiers (Sec. VIII-B). On the usual criteria for secondary analysis of already-public material without intervention we consider the study exempt. The counter-consideration we state rather than resolve in our favour: the images are biometric and the people in them stay identifiable—hence the restrictive release policy. A formal determination has been requested from our university's ethics committee and will be sent to the editor, with its filing date, when issued; it, not our reading of it, governs. *Consent.* Because the data are drawn from public posts with no feasible channel to contact subjects, individual informed consent for participation and for publication was not obtained; we rely on the scientific-research basis and the de-identification safeguards of Sec. VIII-B.

Data were collected from public "then/now" community walls via the platform's official API. We stress that *official API access and voluntary public posting do not by themselves authorize biometric processing.* Under Russia's 152-FZ, the 2020 amendment (519-FZ) replaced "publicly available" data with "data the subject authorized for dissemination" (separate consent required), and Art. 11 requires written consent for biometric data used to identify a person; under the GDPR [22], a face processed for unique identification is special-category data (Art. 9), for which the "manifestly made public" exception is read narrowly (cf. the Clearview AI enforcement actions). We do *not* hold explicit/written consent. We rely on the scientific-research basis (GDPR Art. 6(1)(f) and 9(2)(j) with Art. 89 safeguards; analogous research handling under 152-FZ) and compensate with the minimization, non-redistribution and de-identification measures below. We report this gap transparently rather than claiming full compliance, and we recommend institutional ethics review and data-protection legal counsel before any deployment.

B. De-identification and Release Policy

We *do not release raw face images or raw posts.* The *public* release is restricted to code, configurations, the pseudonymized benchmark protocol and aggregate statistics. Trained weights and derived embeddings are *not* released publicly: a face embedding is itself a comparable *biometric template*, and salted identifier hashes do not render it anonymous, so weights and embeddings are shared with bona-fide researchers *on request* under a brief research-use agreement (research-only; no re-identification; no commercial or surveillance use). The de-identified benchmark dataset is likewise available on request, and formal data-protection documentation will be prepared if a requester or venue requires it. All platform identifiers (owner/photo/post IDs) are replaced by salted hashes; captions/comments (which may contain names) are excluded; only an age bucket and a pseudonymous person ID are kept. Raw crops are retained locally only, under a retention policy. Paper figures contain no identifiable faces (aggregate plots only).

C. DPIA and Minimization

Because the processing is biometric and large-scale, a Data Protection Impact Assessment is conducted and accompanies the release (key risks: re-identification, function creep, minors; mitigations as in Sec. VIII-B and Sec. VIII-D). Data minimization is applied end-to-end (we store a face crop + age + pseudonymous ID, not names or social graphs).

D. Minors

The child↔adult regime necessarily involves images of minors, whose data carry the strictest protection (152-FZ; GDPR Art. 8). Minor-age items (flagged via apparent age) are withheld from any release absent an explicit legal framework and verifiable guardian consent, and are used only for aggregate, non-redistributed analysis.

E. Subject Rights

A public contact and takedown / deletion-on-request procedure are provided; deletion propagates to derived artifacts via a per-item deletion-status field. Where rights are unclear, the item is withheld. A full datasheet [15] and the DPIA accompany the release.

IX. Reproducibility

We release all scripts and configurations; full hyperparameters are in the released configs. Key choices: *usable-face* filtering (det-score, size, Laplacian blur, single target); detector/alignment InsightFace `buffalo_l` RetinaFace + 5-point ArcFace `norm_crop` (112 px; FaceNet input 160 px); ArcFace `w600k_r50` embeddings for *clustering* (union-find, cosine ≥ 0.85) and *dedup* (cosine ≥ 0.97); GigaChat LLM with versioned prompts (cached, resumable); a *by-person* 70/15/15 split (seed 42, balanced negatives, + an age-matched variant); margin-contrastive training (head scope; lr 3e-5 weak / 1e-5 strong; early stop on val-AUC; seeds 42/1/2); pure-NumPy ROC-AUC/EER/TAR@FAR.

Compute. The entire study runs on a single laptop: one NVIDIA RTX 3060 Laptop GPU (6 GB), an AMD Ryzen 7 5800H (8 cores / 16 threads) and 16 GB RAM under Windows 11; the software stack is Python 3.12, PyTorch 2.12 (CUDA 13.2, cuDNN 9.2), with InsightFace 1.0 / ONNX Runtime 1.26 for detection and embeddings and scikit-learn 1.9 / Matplotlib 3.11 for analysis and figures. The weak-backbone protocol keeps the full study within a 6 GB GPU budget.

Data availability. Code, configurations, versioned LLM prompts, frozen splits (by hash) and the de-identified benchmark protocol are released; *raw face images and posts are withheld* (Sec. VIII), so replication of the data-mining stage is *partial*: pseudonymization does not render face data non-personal, and the LLM parser depends on an external versioned service subject to temporal drift, so we release its cached outputs and prompts for deterministic replay rather than a live re-run. The de-identified dataset and derived features are available to bona-fide researchers on request (Sec. VIII).

X. Conclusion and Future Work

Multi-photo posts are a scalable, weakly/naturally-supervised signal of cross-age identity; fine-tuning a weak recognizer on them yields transferable, cross-source, objective-robust gains concentrated on large-gap cross-age matching (without measurable degradation on easy benchmarks, so the gain is concentrated rather than traded off), explained by removing an age shortcut, that are data-efficient and do not reduce AUC in any evaluated apparent stratum. Future work: a larger non-VK longitudinal set (beyond the Reddit cross-platform test of Sec. V-I) and a dedicated child$\leftrightarrow$adult benchmark; a human-audited supervision subset (with human–LLM agreement), resources permitting; per-point multi-seed confidence bands for the headroom curve; a full per-cell multiple-comparison correction; full SOTA add-ons; a real aging model at native resolution; and a calibrated demonstrator under the governance constraints of Sec. VIII.

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